

DATA DICTIONARY DOCUMENT

Shop Sathi Database Dictionary

Complete Tabular Field Reference, Data Types, Constraints & Indexes for all 8 Backend Collections

Collections Summary Index

#	Collection Name	Model Name	Primary Key	Description / Purpose
1	users	User	_id (ObjectId)	Customer and admin accounts, credentials & addresses
2	products	Product	_id (ObjectId)	Product catalog, pricing, inventory, variants & images
3	categories	Category	_id (ObjectId)	Hierarchical product categories and sub-categories
4	carts	Cart	_id (ObjectId)	Active user shopping cart items & applied coupon
5	wishlists	Wishlist	_id (ObjectId)	User saved wishlist items
6	orders	Order	_id (ObjectId)	Placed customer orders, delivery status & financials
7	reviews	Review	_id (ObjectId)	Product ratings and reviews
8	coupons	Coupon	_id (ObjectId)	Promotional discount vouchers and usage tracking

1. `users` Collection Schema

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>_id</code>	ObjectId	Auto	Primary Key	Auto	Unique document identifier
<code>name</code>	String	Yes	No	—	User full name (2–50 chars)
<code>email</code>	String	Yes	Unique Index	—	User email address (lowercase, unique)
<code>password</code>	String	Yes	No	—	Bcrypt hashed password (<code>select: false</code>)
<code>phone</code>	String	No	No	—	Contact mobile number
<code>avatar.url</code>	String	No	No	<code>""</code>	Profile image Cloudinary URL
<code>avatar.publicId</code>	String	No	No	<code>""</code>	Profile image asset public ID
<code>role</code>	String	No	Index	<code>'customer'</code>	Enum: <code>['customer', 'admin', 'superadmin']</code>
<code>addresses</code>	Array[Object]	No	Embedded	<code>[]</code>	Embedded address sub-documents
<code>addresses._id</code>	ObjectId	Yes	Sub-ID	Auto	Unique address identifier
<code>addresses.label</code>	String	No	No	—	Label (e.g. "Home", "Work", "Office")
<code>addresses.fullName</code>	String	Yes	No	—	Recipient full name
<code>addresses.phone</code>	String	Yes	No	—	Contact phone number
<code>addresses.street</code>	String	Yes	No	—	Street / Building address
<code>addresses.city</code>	String	Yes	No	—	City / Town
<code>addresses.state</code>	String	Yes	No	—	State / Province
<code>addresses.pincode</code>	String	Yes	No	—	Postal pincode

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>addresses.country</code>	String	No	No	<code>'India'</code>	Country name
<code>addresses.isDefault</code>	Boolean	No	No	<code>false</code>	Default shipping address flag
<code>isBlocked</code>	Boolean	No	Index	<code>false</code>	Account blocked status flag
<code>resetPasswordToken</code>	String	No	No	—	SHA-256 password reset token
<code>resetPasswordExpire</code>	Date	No	No	—	Password reset token expiration date
<code>createdAt</code>	Date	Auto	No	Current Date	Account creation timestamp
<code>updatedAt</code>	Date	Auto	No	Current Date	Last account modification timestamp

2. `products` Collection Schema

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>_id</code>	ObjectId	Auto	Primary Key	Auto	Unique product identifier
<code>name</code>	String	Yes	Text Index	—	Product title (3–200 chars)
<code>slug</code>	String	Auto	Unique Index	Auto	URL slug generated from name
<code>description</code>	String	Yes	Text Index	—	Full detailed description
<code>shortDescription</code>	String	No	No	—	Short overview (Max 300 chars)
<code>price</code>	Number	Yes	Index	—	Current selling price (≥ 0)
<code>originalPrice</code>	Number	No	No	—	Strike-through MRP (≥ 0)
<code>discount</code>	Number	Auto	No	<code>0</code>	Calculated discount percentage (0–100%)
<code>category</code>	ObjectId	Yes	Foreign Key	—	Reference to <code>Category._id</code>
<code>brand</code>	String	No	Text Index	—	Brand / Manufacturer name
<code>sku</code>	String	No	Sparse Unique	—	Stock Keeping Unit (Uppercase)
<code>images</code>	Array[Object]	No	Embedded	<code>[]</code>	Array of product image objects
<code>images._id</code>	ObjectId	Yes	Sub-ID	Auto	Image sub-document ID
<code>images.url</code>	String	Yes	No	—	Cloudinary image URL
<code>images.publicId</code>	String	Yes	No	—	Cloudinary asset public ID
<code>images.isMain</code>	Boolean	No	No	<code>false</code>	Main cover image flag
<code>variants</code>	Array[Object]	No	Embedded	<code>[]</code>	Product variant options (Size, Color)

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
variants.name	String	Yes	No	—	Variant group (e.g. "Size")
variants.options	Array[String]	Yes	No	—	Option values (e.g. ["S", "M", "L"])
specifications	Array[Object]	No	Embedded	[]	Key-value specifications
specifications.key	String	Yes	No	—	Spec attribute name (e.g. "Material")
specifications.value	String	Yes	No	—	Spec attribute value (e.g. "Cotton")
tags	Array[String]	No	Text Index	[]	Search tags
stock	Number	Yes	Index	0	Available stock count (>= 0)
isActive	Boolean	No	Index	true	Active status flag (Soft delete)
isFeatured	Boolean	No	Index	false	Show on homepage banner
ratings.average	Number	Auto	Index	0	Average star rating (0–5)
ratings.count	Number	Auto	No	0	Total review count
createdBy	ObjectId	No	Foreign Key	—	Reference to User._id (Creator)
createdAt	Date	Auto	No	Current Date	Creation timestamp
updatedAt	Date	Auto	No	Current Date	Last modification timestamp

3. `categories` Collection Schema

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>_id</code>	ObjectId	Auto	Primary Key	Auto	Unique category identifier
<code>name</code>	String	Yes	Unique Index	—	Category title (2–50 chars)
<code>slug</code>	String	Auto	Unique Index	Auto	URL slug generated from name
<code>description</code>	String	No	No	—	Category description
<code>image.url</code>	String	No	No	<code>""</code>	Banner image URL
<code>image.publicId</code>	String	No	No	<code>""</code>	Cloudinary asset ID
<code>parent</code>	ObjectId	No	Foreign Key	<code>null</code>	Reference to parent <code>Category._id</code> (null = main)
<code>isActive</code>	Boolean	No	Index	<code>true</code>	Active status flag
<code>createdAt</code>	Date	Auto	No	Current Date	Creation timestamp
<code>updatedAt</code>	Date	Auto	No	Current Date	Modification timestamp

4. `carts` Collection Schema

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>_id</code>	ObjectId	Auto	Primary Key	Auto	Unique cart identifier
<code>user</code>	ObjectId	Yes	Unique FK	—	Reference to <code>User._id</code> (One cart per user)
<code>items</code>	Array[Object]	No	Embedded	<code>[]</code>	Cart line items
<code>items._id</code>	ObjectId	Yes	Sub-ID	Auto	Cart item ID
<code>items.product</code>	ObjectId	Yes	Foreign Key	—	Reference to <code>Product._id</code>
<code>items.name</code>	String	Yes	No	—	Product name snapshot
<code>items.image</code>	String	No	No	<code>""</code>	Image URL snapshot
<code>items.price</code>	Number	Yes	No	—	Unit price snapshot
<code>items.quantity</code>	Number	Yes	No	<code>1</code>	Quantity (≥ 1)
<code>items.selectedVariants</code>	Map[String]	No	No	—	Selected variant key-values
<code>coupon</code>	ObjectId	No	Foreign Key	<code>null</code>	Reference to applied <code>Coupon._id</code>
<code>discountAmount</code>	Number	No	No	<code>0</code>	Applied discount amount
<code>createdAt</code>	Date	Auto	No	Current Date	Cart creation timestamp
<code>updatedAt</code>	Date	Auto	No	Current Date	Last item change timestamp

5. `wishlists` Collection Schema

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>_id</code>	ObjectId	Auto	Primary Key	Auto	Unique wishlist identifier
<code>user</code>	ObjectId	Yes	Unique FK	—	Reference to <code>User._id</code> (One wishlist per user)
<code>products</code>	Array[ObjectId]	No	Array FK	<code>[]</code>	Array of <code>Product._id</code> references
<code>createdAt</code>	Date	Auto	No	Current Date	Creation timestamp
<code>updatedAt</code>	Date	Auto	No	Current Date	Last change timestamp

6. `orders` Collection Schema

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>_id</code>	ObjectId	Auto	Primary Key	Auto	Unique order ID
<code>orderNumber</code>	String	Auto	Unique Index	Auto	Format: <code>ORD-YYYYMMDD-XXXX</code>
<code>user</code>	ObjectId	Yes	Foreign Key	—	Reference to <code>User._id</code>
<code>items</code>	Array[Object]	Yes	Embedded	—	Order line items snapshot
<code>items.product</code>	ObjectId	Yes	Foreign Key	—	Reference to <code>Product._id</code>
<code>items.name</code>	String	Yes	No	—	Product title snapshot
<code>items.image</code>	String	No	No	<code>""</code>	Thumbnail snapshot
<code>items.price</code>	Number	Yes	No	—	Unit price snapshot
<code>items.quantity</code>	Number	Yes	No	—	Ordered quantity (≥ 1)
<code>items.selectedVariants</code>	Map[String]	No	No	—	Variant options snapshot
<code>shippingAddress.fullName</code>	String	Yes	No	—	Recipient full name
<code>shippingAddress.phone</code>	String	Yes	No	—	Recipient phone number
<code>shippingAddress.street</code>	String	Yes	No	—	Street address
<code>shippingAddress.city</code>	String	Yes	No	—	City
<code>shippingAddress.state</code>	String	Yes	No	—	State
<code>shippingAddress.pincode</code>	String	Yes	No	—	Postal pincode
<code>shippingAddress.country</code>	String	No	No	<code>'India'</code>	Country

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
paymentMethod	String	Yes	No	—	Enum: ['COD', 'ONLINE']
paymentStatus	String	No	Index	'pending'	Enum: ['pending', 'paid', 'failed', 'refunded']
orderStatus	String	No	Index	'pending'	Enum: ['pending', 'processing', 'shipped', 'delivered', 'cancelled']
statusHistory	Array[Object]	No	Embedded	[]	Audit trail of status updates
coupon	ObjectId	No	Foreign Key	null	Reference to Coupon._id
subtotal	Number	Yes	No	—	Items price subtotal
shippingCharge	Number	No	No	0	Shipping charge (Free above ₹499)
taxAmount	Number	No	No	0	GST tax (18%)
discountAmount	Number	No	No	0	Discount deduction
totalAmount	Number	Yes	No	—	Final payable total
estimatedDelivery	Date	No	No	—	Expected delivery date
deliveredAt	Date	No	No	—	Delivery completion date
cancelledAt	Date	No	No	—	Order cancellation date
cancelReason	String	No	No	—	Reason for cancellation
createdAt	Date	Auto	Index	Current Date	Order date
updatedAt	Date	Auto	No	Current Date	Last status change date

7. `reviews` Collection Schema

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>_id</code>	ObjectId	Auto	Primary Key	Auto	Unique review ID
<code>product</code>	ObjectId	Yes	Compound Unique	—	Reference to <code>Product._id</code>
<code>user</code>	ObjectId	Yes	Compound Unique	—	Reference to <code>User._id</code>
<code>rating</code>	Number	Yes	No	—	Star rating score (1 to 5)
<code>title</code>	String	No	No	—	Review title (Max 100 chars)
<code>comment</code>	String	Yes	No	—	Review text (10–1000 chars)
<code>isVerifiedPurchase</code>	Boolean	No	No	<code>false</code>	True if verified purchase
<code>createdAt</code>	Date	Auto	Index	Current Date	Review submission date
<code>updatedAt</code>	Date	Auto	No	Current Date	Last edit date

8. coupons Collection Schema

Field Name	Data Type	Required	Key / Index	Default Value	Description / Constraints
<code>_id</code>	ObjectId	Auto	Primary Key	Auto	Unique coupon ID
<code>code</code>	String	Yes	Unique Index	—	Coupon code (3–20 chars, uppercase)
<code>discountType</code>	String	Yes	No	—	Enum: <code>['percentage', 'flat']</code>
<code>discountValue</code>	Number	Yes	No	—	Percentage % or flat amount ₹
<code>minOrderAmount</code>	Number	No	No	<code>0</code>	Minimum order subtotal required
<code>maxDiscount</code>	Number	No	No	<code>null</code>	Max discount cap for percentage
<code>usagelimit</code>	Number	No	No	<code>null</code>	Total max redemptions (null = unlimited)
<code>usedCount</code>	Number	No	No	<code>0</code>	Current total redemptions
<code>perUserLimit</code>	Number	No	No	<code>1</code>	Max redemptions per user
<code>usedBy</code>	Array[ObjectId]	No	Foreign Keys	<code>[]</code>	Array of <code>User._id</code> references
<code>isActive</code>	Boolean	No	Index	<code>true</code>	Active status flag
<code>expiresAt</code>	Date	Yes	Index	—	Expiration timestamp
<code>createdAt</code>	Date	Auto	No	Current Date	Creation timestamp
<code>updatedAt</code>	Date	Auto	No	Current Date	Modification timestamp

Database Indexes Reference Table

Collection	Index Fields	Index Type	Unique	Purpose
users	{ email: 1 }	B-Tree	Yes	Fast unique user login lookup
users	{ role: 1 }	B-Tree	No	Role based user filtering
users	{ isBlocked: 1 }	B-Tree	No	Account status filter
products	{ slug: 1 }	B-Tree	Yes	Fast product detail URL lookup
products	{ category: 1 }	B-Tree	No	Category product listing filter
products	{ isActive: 1, isFeatured: 1 }	Compound	No	Homepage featured products query
products	{ isActive: 1, price: 1 }	Compound	No	Price sorting and filtering
products	{ 'ratings.average': -1 }	B-Tree	No	Top rated products sorting
products	{ name: 'text', description: 'text', tags: 'text', brand: 'text' }	Text	No	Full-text search search bar
categories	{ slug: 1 }	B-Tree	Yes	Category page URL lookup
categories	{ parent: 1 }	B-Tree	No	Sub-category hierarchy queries
orders	{ orderNumber: 1 }	B-Tree	Yes	Order tracking code lookup
orders	{ user: 1, createdAt: -1 }	Compound	No	Customer order history listing
orders	{ orderStatus: 1 }	B-Tree	No	Admin order fulfillment filter
reviews	{ product: 1, user: 1 }	Compound	Yes	Prevent duplicate user reviews per product
reviews	{ product: 1, createdAt: -1 }	Compound	No	Product review list pagination
coupons	{ code: 1 }	B-Tree	Yes	Fast coupon code validation

Collection	Index Fields	Index Type	Unique	Purpose
coupons	{ isActive: 1, expiresAt: 1 }	Compound	No	Active coupon validity queries
carts	{ user: 1 }	B-Tree	Yes	Single cart per user enforcement
wishlists	{ user: 1 }	B-Tree	Yes	Single wishlist per user enforcement